Stress, elastic constants and the equation of state from the FOCEX force constants

FOCEX notes (draft for checking)

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Abstract

Part I gives the Huang conditions on the harmonic force constants (FCs) of a reference structure under stress, as now imposed in FOCEX. Part II derives, from FC1–FC4, the bulk modulus B_0, its pressure derivative B_0' and the third-order elastic constants, for any number of atoms per cell and any symmetry. It then gives the route to the finite-temperature equation of state: $V(T)$, $\alpha(T)$, $C_P(T)$, $B_T(T)$. It also corrects the B_0' formula of the ALATDYN paper and of FOCEX's `cubic_elastic`.

1 Notation (as in the ALATDYN paper)

- Atom τ of the cell at lattice vector R sits at $X_{R\tau} = R + \tau$. Atom labels are $i = (R\tau)$, N is the number of cells and V_0 the volume of the primitive cell. Greek letters are Cartesian indices, summed when repeated.
- $\mathrm{Tr}\, M$ and $\mathrm{Det}\, M$ are the trace and the determinant of a 3×3 matrix M; $\mathbb{1}$ is the unit matrix.
- The potential energy is expanded in the displacements $u_{i\alpha}$:

$$V^{LD} = E_0 + \sum_i \Pi_i u_i + \frac{1}{2!}\sum_{ij} \Phi_{ij} u_i u_j + \frac{1}{3!}\sum_{ijk} \Psi_{ijk} u_i u_j u_k + \frac{1}{4!}\sum_{ijkl} \chi_{ijkl} u_i u_j u_k u_l. \quad (1)$$

  All FCs obey the acoustic sum rule (ASR) in each index, e.g. $\sum_k \Psi_{ijk} = 0$.
- A homogeneous deformation is described by the displacement gradient $\eta_{\alpha\beta} = \partial u_\alpha / \partial r_\beta$ and the internal displacements u^0_τ. These are *independent* variables. The atoms move by

$$S_{R\tau\alpha} = \eta_{\alpha\beta}(R + \tau)_\beta + u^0_{\tau\alpha}. \quad (2)$$

- Any deformation is a rotation times a symmetric stretch (polar decomposition), and the energy does not depend on the rotation. In Part II we therefore take η symmetric. It then equals the Cauchy strain ε, and has 6 components. In Part I, η is general.

The harmonic coefficients are those of the paper:

$$\phi_{\tau\tau'} = \frac{1}{N}\sum_{RR'} \Phi_{R\tau,R'\tau'}, \qquad \Gamma = \phi^+ \ \text{(pseudo-inverse; the 3 translations removed)},$$
$$Q_{\tau\alpha;\alpha'\beta'} = \frac{1}{N}\sum_{RR'\tau'} \Phi^{\alpha\alpha'}_{R\tau,R'\tau'}\,(R' + \tau' - R - \tau)_{\beta'}, \quad (3)$$
$$A_{\alpha\beta,\alpha'\beta'} = -\frac{1}{2}\sum_{\tau}\sum_{R'\tau'} \Phi^{\alpha\alpha'}_{0\tau,R'\tau'}\, d_\beta d_{\beta'}, \qquad d = R' + \tau' - \tau.$$

In $A_{\alpha\beta,\alpha'\beta'}$, α, α' label displacements and β, β' coordinates.

Absolute versus relative sums. The paper also writes A and Q with absolute positions, e.g. $A = \frac{1}{N}\sum \Phi(R+\tau)(R'+\tau')$. For a finite sum the two forms agree by the ASR. But for the infinite crystal the absolute sums are only conditionally convergent, and their value depends on the image convention (Sec. 5). We use the relative forms of Eq. (3) throughout. That these give the true second derivatives of the energy of a homogeneously deformed crystal is the assumption behind Eq. (8). The numerical check of Sec. 4 tests it.

2 Energy of a homogeneous deformation

Inserting Eq. (2) into Eq. (1) gives the energy per cell as a polynomial in $q = (\eta, u^0)$, because S is linear in q:

$$W(\eta, u^0) = W_0 + W_1(q) + \tfrac{1}{2}W_2(q,q) + \tfrac{1}{6}W^{(3)}(q,q,q) + \tfrac{1}{24}W^{(4)}(q,q,q,q). \tag{4}$$

W_n is FCn contracted with n copies of $S(q)$, per cell, evaluated with relative vectors. In particular $W_2(q,q) = q^{\mathsf{T}} H q$, with

$$H = \begin{pmatrix} A & Q^{\mathsf{T}} \\ Q & \phi \end{pmatrix}. \tag{5}$$

The linear term has no relative form. From the FCs alone it would be $\Xi : \eta + \Pi \cdot u^0$, which is image-dependent and vanishes for $\Pi = 0$ even when the crystal is stressed (Sec. 5). We therefore take $W_1 = V_0\sigma : \eta + \Pi \cdot u^0$, with σ the input stress of the reference. $W^{(3)}$ and $W^{(4)}$ are computed in Sec. 6.

The elastic energy is obtained in two steps. The internal displacements adjust to the strain. So we first minimise over u^0 at fixed η, and only then differentiate with respect to η:

$$W_{\mathrm{rel}}(\eta) = \min_{u^0} W(\eta, u^0). \tag{6}$$

We assume $\Pi = 0$, i.e. the atoms are at rest in the reference. If not, relax them first, $u^0 = -\Gamma\Pi$, and re-expand the FCs there, Eq. (24). To second order, $\partial W/\partial u^0 = Q\eta + \phi u^0 = 0$ gives

$$u_1^0(\eta) = -\Gamma Q \eta, \qquad C = \frac{1}{V_0}\frac{\partial^2 W_{\mathrm{rel}}}{\partial\eta\,\partial\eta} = \frac{1}{V_0}\,\mathrm{Sym}\big(A - Q^{\mathsf{T}}\Gamma Q\big), \tag{7}$$

the elastic tensor of the paper. "Sym" symmetrises over $\alpha \leftrightarrow \beta$ within each pair, because η is symmetric.

I. Huang conditions for a stressed reference

3 Rotational invariance and the stress

Here η is general. A rigid rotation costs no energy, so E depends on the deformation only through the Lagrangian strain $\varepsilon^L_{\alpha\gamma} = \frac{1}{2}(\eta_{\alpha\gamma} + \eta_{\gamma\alpha} + \eta_{\theta\alpha}\eta_{\theta\gamma})$. Define the stress of the reference $\sigma = V^{-1}\partial E/\partial\varepsilon^L$ and $c = V^{-1}\partial^2 E/\partial\varepsilon^L\partial\varepsilon^L$ (symmetric in each pair and under exchanging the pairs). Then

$$\frac{E}{V} = \frac{E_0}{V} + \sigma_{\alpha\gamma}\varepsilon^L_{\alpha\gamma} + \tfrac{1}{2}c_{\alpha\gamma\beta\lambda}\varepsilon^L_{\alpha\gamma}\varepsilon^L_{\beta\lambda} + \dots$$

The part of second order in η contains, besides $\frac{1}{2}c\,\eta\eta$, the term $\frac{1}{2}\sigma_{\gamma\lambda}\eta_{\theta\gamma}\eta_{\theta\lambda}$ from the quadratic part of ε^L. Compare it with W_{rel} to second order, $\frac{1}{2}(A - Q^{\mathsf{T}}\Gamma Q)\eta\eta$ for general η. The coefficient of $\eta_{\alpha\gamma}\eta_{\beta\lambda}$ gives

$$\frac{1}{V_0}\big(A_{\alpha\gamma,\beta\lambda} - (Q^{\mathsf{T}}\Gamma Q)_{\alpha\gamma,\beta\lambda}\big) = c_{\alpha\gamma\beta\lambda} + \delta_{\alpha\beta}\,\sigma_{\gamma\lambda}. \tag{8}$$

The rank-2 rotational invariance (Leibfried–Ludwig) reads, with $e_{\beta\gamma\nu}$ the Levi-Civita symbol,

$$\sum_{R'\tau'} \Phi^{\alpha\beta}_{0\tau,R'\tau'} X_{R'\tau',\gamma}\, e_{\beta\gamma\nu} + \Pi_{\tau\beta}\, e_{\beta\alpha\nu} = 0.$$

For $\Pi = 0$ it makes $Q_{\tau\theta;\beta\lambda}$ symmetric in (β, λ). So $Q^{\mathsf{T}}\Gamma Q$ is symmetric under $\alpha \leftrightarrow \gamma$ and under $\beta \leftrightarrow \lambda$, separately.

4 Generalised Huang conditions

Exchange $\alpha \leftrightarrow \gamma$ and $\beta \leftrightarrow \lambda$ in Eq. (8). c and $Q^{\mathsf{T}}\Gamma Q$ do not change, and $A_{\alpha\gamma,\beta\lambda}$ becomes $A_{\gamma\alpha,\lambda\beta}$. Subtracting the two equations:

$$\boxed{A_{\alpha\gamma,\beta\lambda} - A_{\gamma\alpha,\lambda\beta} = V_0\big(\delta_{\alpha\beta}\,\sigma_{\gamma\lambda} - \delta_{\gamma\lambda}\,\sigma_{\alpha\beta}\big)} \tag{9}$$

- For $\sigma = 0$ these are Huang's conditions [1].
- For a hydrostatic stress, $\sigma = -P\mathbb{1}$, the right-hand side is $-P(\delta_{\alpha\beta}\delta_{\gamma\lambda} - \delta_{\gamma\lambda}\delta_{\alpha\beta}) = 0$. So FC2 determines the deviatoric stress through Eq. (9), but not the pressure.
- With Voigt pairs there are 15 conditions, as for $\sigma = 0$.
- The derivation assumes $\Pi = 0$. The exact-model test below finds the relation satisfied also with $\Pi \neq 0$; we have not derived why.

Verification. We used the exact FC2 of a pair potential $V(r) = A(r_c^2 - r^2)^2 + B(r_c^2 - r^2)^3$, with the exact stress $\sigma_{\alpha\beta} = \frac{1}{2V_0}\sum_{\tau,j} V'(r)\, d_\alpha d_\beta / r$. The crystals were the seven crystal systems, plus diamond and hcp ($\Pi = 0$). Eq. (9) holds to 3×10^{-13} (model units) in all nine cases, with $|\sigma|$ up to 529. The exact-model test of FOCEX (`FOCEX/REFERENCE/exact_model`) adds wurtzite ZnO and monoclinic HfO_2, which have $\Pi \neq 0$: the right-hand side of the FOCEX row below is up to 0.23 and 0.43 eV, and the relation holds to 2×10^{-13} eV.

In FOCEX (since 2026-10-26). For Voigt pairs $(\alpha\beta)$, $(\gamma\delta)$ the Huang row is

$$\sum_{\tau}\sum_{R'\tau'} \Big[\Phi^{\alpha\beta}_{0\tau,R'\tau'} d_\gamma d_\delta - \Phi^{\gamma\delta}_{0\tau,R'\tau'} d_\alpha d_\beta\Big] = -2V_0\big(\delta_{\alpha\beta}\sigma_{\gamma\delta} - \delta_{\gamma\delta}\sigma_{\alpha\beta}\big).$$

σ is read from the optional file `stress.params` (3×3 in GPa, tensile positive; zero if the file is absent). Because of the right-hand side, the constrained fit is $x = x_p + Ky$. *Still to be checked: the sign convention of the stress printed by VASP.*

5 The stress is an input

The paper's residual stress is $V_0\sigma = \mathrm{Sym}(\Xi - \Pi\Gamma Q)$, with $\Xi_{\alpha\beta} = \sum_\tau \Pi_{\tau\alpha}\tau_\beta$. When $\Pi \neq 0$ it depends on which periodic image of each atom is taken as τ:

- replacing τ by $\tau + R$ changes Ξ by $\Pi_\tau R$;
- Q does not change, because its relative form in Eq. (3) sums over all R.

When $\Pi = 0$ the formula gives zero, whatever the stress. FC2 fixes only the deviatoric stress (Sec. 4). So the stress, and hence the linear term $V_0\sigma : \eta$ of W, is an *input*: the DFT stress of the reference, the number in `stress.params`. The rank-1 rotational rows were dropped for the same reason: they too depend on the image choice when $\Pi \neq 0$.

II. Equation of state

6 The relaxed energy to fourth order

6.1 Expansion of $W_{\rm rel}$

Write the relaxed internal displacements as $u^0(\eta) = u_1^0(\eta) + u_2^0(\eta,\eta) + \dots$, and let $p(\eta) = (\eta, u_1^0(\eta))$ be the linearly relaxed deformation. Then

$$W_{\rm rel}(\eta) = W_0 + V_0\sigma : \eta + \tfrac{1}{2}V_0\, C(\eta,\eta) + \tfrac{1}{6}W_{\rm rel}^{(3)}(\eta,\eta,\eta) + \tfrac{1}{24}W_{\rm rel}^{(4)}(\eta,\eta,\eta,\eta), \tag{10}$$

with

$$W_{\rm rel}^{(3)}(\eta,\eta,\eta) = W^{(3)}(p,p,p), \tag{11}$$

$$W_{\rm rel}^{(4)}(\eta,\eta,\eta,\eta) = W^{(4)}(p,p,p,p) - 3\, w^{\mathsf T}\Gamma\, w, \qquad w_{\tau\alpha} = W^{(3)}(p,p,e_{\tau\alpha}). \tag{12}$$

Here $e_{\tau\alpha}$ is the unit displacement of atom τ (all cells) in direction α.

Derivation. Put $u^0 = u_1^0 + \delta$ in Eq. (4), with $\delta = O(\eta^2)$.

- At $\delta = 0$ the force on the atoms is $-\partial W/\partial u^0 = -(Q\eta + \phi u_1^0) - \frac{1}{2}w + O(\eta^3) = -\frac{1}{2}w + O(\eta^3)$, because u_1^0 cancels the first-order force.
- The terms in δ are therefore $\frac{1}{2}w\cdot\delta + \frac{1}{2}\delta^{\mathsf T}\phi\,\delta + O(\eta^5)$. Minimising gives $\delta = u_2^0 = -\frac{1}{2}\Gamma w$ and the energy $-\frac{1}{8}w^{\mathsf T}\Gamma w$, of order η^4.
- So the cubic term of $W_{\rm rel}$ needs only u_1^0 (the "$2n+1$" rule). The quartic term needs u_2^0, through $-\frac{1}{8}w^{\mathsf T}\Gamma w = \frac{1}{24}(-3w^{\mathsf T}\Gamma w)$.

Γ excludes the translations. That is consistent, because $\sum_\tau w_\tau = 0$: the internal forces of the crystal sum to zero, and all cells are equivalent.

6.2 How to compute $W^{(3)}$ and $W^{(4)}$

- **The Hessian at a deformed point.** At $q = \lambda p$, the FC2 of the crystal is (Eq. (24))

  $$\Phi_{ij}(\lambda p) = \Phi_{ij} + \lambda\sum_k \Psi_{ijk}S_k + \frac{\lambda^2}{2}\sum_{kl}\chi_{ijkl}S_kS_l, \qquad S = S(p).$$

  By the ASR, S_k can be replaced by $S_k - S_i$, and S_kS_l by $(S_k - S_i)(S_l - S_i)$.

- **No deformed vectors are needed.** Consider a further deformation η' of the displaced crystal, written in the reference coordinates, $S' = \eta' X$. Its relative vectors are $\eta' d$, whatever λ. So the Hessian of W at λp is $H[\Phi(\lambda p)]$, the blocks of Eq. (3) built with $\Phi(\lambda p)$ and the *reference* vectors d.

- **The result.** Since $\frac{d}{d\lambda}\, p^{\mathsf T}\nabla^2 W(\lambda p)\, p = W^{(3)}(p,p,p)$ and similarly at second order,

  $$W^{(3)}(p,p,p) = p^{\mathsf T}H[\Psi S]\, p, \qquad W^{(4)}(p,p,p,p) = p^{\mathsf T}H[\chi SS]\, p. \tag{13}$$

  Here $H[\Delta\Phi]$ means the blocks evaluated with $\Delta\Phi$ in place of Φ. Numerical check: differences of C at $\pm\lambda$.

- **The vector** w of Eq. (12) is $w_\tau = \sum_{jk}\Psi_{\tau jk}(S_j - S_\tau)(S_k - S_\tau)$, with τ in cell 0 and j,k over all atoms.

- **Images.** Choosing another image of an atom, $\tau \to \tau + R$, changes u_1^0 by $-\eta R$ and leaves S unchanged. So the results do not depend on that choice.

7 Bulk modulus and its pressure derivative at $T = 0$

7.1 Definitions

$B = -V\,dP/dV$ and $B' = dB/dP$. With $P = -E'(V)$ we have $dB/dV = E'' + VE'''$ and $dP/dV = -E''$, so

$$B' = -1 - V\,\frac{E'''}{E''}. \tag{14}$$

Check: the Murnaghan equation $P = (B_0/B')[(V_0/V)^{B'} - 1]$ gives $V_0E'''/E'' = -(1 + B')$.

7.2 Expansion of the enthalpy

Under a pressure P the crystal minimises the enthalpy per cell, $G(\eta) = W_{\rm rel}(\eta) + P\,V(\eta)$, over the 6 strains. The internal coordinates are already relaxed in $W_{\rm rel}$. For a symmetric stretch the volume is exactly

$$\frac{V(\eta)}{V_0} = \mathrm{Det}(\mathbb{1} + \eta) = 1 + \mathrm{Tr}\,\eta + \tfrac{1}{2}\big[(\mathrm{Tr}\,\eta)^2 - \mathrm{Tr}(\eta^2)\big] + \mathrm{Det}\,\eta, \tag{15}$$

the characteristic-polynomial identity of a 3×3 matrix. We write it as

$$V = V_0 + v \cdot \eta + \tfrac{1}{2}V_2(\eta,\eta) + \tfrac{1}{6}V_3(\eta,\eta,\eta),$$

with $v_{\alpha\beta} = V_0\delta_{\alpha\beta}$, $V_2(\eta,\eta) = V_0[(\mathrm{Tr}\,\eta)^2 - \mathrm{Tr}(\eta^2)]$ and $V_3(\eta,\eta,\eta) = 6V_0\,\mathrm{Det}\,\eta$.

Let the reference be in equilibrium at a hydrostatic pressure P_0: $\Pi = 0$ and $\sigma = -P_0\mathbb{1}$, so $V_0\sigma : \eta = -P_0\,v \cdot \eta$. Put $P = P_0 + p$ and $\eta = \eta_1 p + \eta_2 p^2 + \dots$ into $\partial G/\partial\eta = 0$. Order by order in p:

$$O(p): \quad \tilde{H}\eta_1 + v = 0, \qquad \tilde{H} = V_0C + P_0V_2, \tag{16}$$

$$O(p^2): \quad \tilde{H}\eta_2 + \tfrac{1}{2}\tilde{W}^{(3)}(\eta_1,\eta_1) + V_2\eta_1 = 0, \qquad \tilde{W}^{(3)} = W^{(3)}_{\rm rel} + P_0V_3. \tag{17}$$

$\eta_1 = -\tilde{H}^{-1}v$ is the strain per unit pressure. For $P_0 = 0$ it is $\eta_1 = -S\mathbb{1}$, with $S = C^{-1}$ the compliance: the cell shape relaxes, as it must in an anisotropic crystal. The volume is $V = V_0(1 + ap + bp^2)$, with

$$V_0a = v \cdot \eta_1, \qquad V_0b = v \cdot \eta_2 + \tfrac{1}{2}V_2(\eta_1,\eta_1).$$

Only η_1 is needed. Since $v^{\mathsf{T}} = -\eta_1^{\mathsf{T}}\tilde{H}$, the $O(p^2)$ equation gives $v \cdot \eta_2 = \frac{1}{2}\tilde{W}^{(3)}(\eta_1,\eta_1,\eta_1) + V_2(\eta_1,\eta_1)$. From $B = -V\,dp/dV$ at $p = 0$, $B = -1/a$ and $B' = 2bB^2 - 1$. Therefore

$$\boxed{B = \frac{V_0}{v^{\mathsf{T}}\tilde{H}^{-1}v}, \qquad B' = \frac{B^2}{V_0}\Big[\tilde{W}^{(3)}(\eta_1,\eta_1,\eta_1) + 3V_2(\eta_1,\eta_1)\Big] - 1} \tag{18}$$

where $\tilde{W}^{(3)}(\eta_1,\eta_1,\eta_1) = W^{(3)}(p_1,p_1,p_1) + 6P_0V_0\,\mathrm{Det}\,\eta_1$ and $p_1 = (\eta_1, -\Gamma Q\eta_1)$.

- For $P_0 = 0$, $B = 1/\sum_{i,j\le3} S_{ij}$, the Reuss value FOCEX reports.
- The cell-shape and internal relaxations enter only through η_1 and p_1. Nothing else in the formula depends on the crystal system.
- The second-order relaxations η_2 and u_2^0 do not appear. $W^{(4)}$ does not enter B'; it enters B''.

Check: cubic crystal with $u_1^0 = 0$. Take $P_0 = 0$. Then $\eta_1 = -\mathbb{K}/(3B)$, and

$$3V_2(\eta_1, \eta_1) = 3V_0\Big[\frac{1}{B^2} - \frac{1}{3B^2}\Big] = \frac{2V_0}{B^2}.$$

Let $f(\varepsilon)$ be the energy per cell under $\eta = \varepsilon\mathbb{K}$. Then $W^{(3)}(p_1, p_1, p_1) = f'''\,(-\frac{1}{3B})^3$, and Eq. (18) gives

$$B_0' = 1 - \frac{f'''}{27V_0B_0}, \qquad f'' = 9V_0B_0. \tag{19}$$

The same follows from Eq. (14) with $V = V_0(1+\varepsilon)^3$ and the chain rule. The "1" comes from $V(\varepsilon)$ being cubic in ε.

8 Third-order elastic constants

Define the third-order elastic constants (with respect to the stretch η) as the coefficients of the cubic term of the relaxed energy, fully symmetric:

$$W_{\mathrm{rel}}^{(3)}(\eta, \eta, \eta) = V_0\, C_{\alpha\beta\gamma\delta\mu\nu}\, \eta_{\alpha\beta}\eta_{\gamma\delta}\eta_{\mu\nu}. \tag{20}$$

Write $\eta = \eta_{\alpha\beta}e^{(\alpha\beta)}$, with $e^{(\alpha\beta)}_{\gamma\delta} = \frac{1}{2}(\delta_{\alpha\gamma}\delta_{\beta\delta} + \delta_{\alpha\delta}\delta_{\beta\gamma})$, and use Eq. (11). Since $W^{(3)}$ is trilinear and p linear in η,

$$C_{\alpha\beta\gamma\delta\mu\nu} = \frac{1}{V_0}\, W^{(3)}\big(p^{(\alpha\beta)}, p^{(\gamma\delta)}, p^{(\mu\nu)}\big), \qquad p^{(\alpha\beta)} = \big(e^{(\alpha\beta)},\, -\Gamma Q\, e^{(\alpha\beta)}\big). \tag{21}$$

- $C_{\alpha\beta\gamma\delta\mu\nu}$ is symmetric within each pair and under any permutation of the pairs: at most 56 independent components (triclinic).
- To compute it, use Eq. (13) with three arguments, $W^{(3)}(p, p', p'') = p'^{\mathsf{T}} H[\Psi S(p)]\, p''$.
- B_0' needs only one contraction: $W^{(3)}(p_1, p_1, p_1) = V_0 C_{\alpha\beta\gamma\delta\mu\nu}\, \eta_{1,\alpha\beta}\, \eta_{1,\gamma\delta}\, \eta_{1,\mu\nu}$.

Lagrangian (Brugger) constants. Tabulated third-order constants are usually defined with respect to the Lagrangian strain, $\varepsilon^L = \eta + \frac{1}{2}\eta^2$ for a symmetric stretch. Insert ε^L into $V_0\big[\sigma : \varepsilon^L + \frac{1}{2}c(\varepsilon^L, \varepsilon^L) + \frac{1}{6}c^{(3)}(\varepsilon^L, \varepsilon^L, \varepsilon^L)\big]$. The terms of third order in η are $V_0[\frac{1}{6}c^{(3)}(\eta, \eta, \eta) + \frac{1}{2}c(\eta, \eta^2)]$; the stress term has none. Comparing with $\frac{1}{6}W_{\mathrm{rel}}^{(3)}$ gives the identity of cubic forms

$$c^{(3)}(\eta, \eta, \eta) = \frac{1}{V_0}W_{\mathrm{rel}}^{(3)}(\eta, \eta, \eta) - 3\, c(\eta, \eta^2). \tag{22}$$

The components follow by polarisation. c is the second-order constant of Eq. (8); the quadratic terms give $C(\eta, \eta) = c(\eta, \eta) + \sigma : \eta^2$, so $c = C$ for a stress-free reference.

8.1 The B_0' formula in the paper and in FOCEX

The paper (section "Thermodynamic properties within the QHA") writes

$$B_0' = V_0\frac{E'''}{E''} = \frac{1}{27V_0B}\frac{d^3E_0}{d\varepsilon^3}, \qquad \frac{d^3E_0}{d\varepsilon^3} = \frac{1}{6N}\sum \Psi X_1X_2X_3 \quad (u^0 = 0).$$

Four corrections:

1. The sign is wrong and the 1 is missing: compare Eq. (14) and Eq. (19).
2. With the 1/3! of Eq. (1), the cubic term of the energy is $\frac{\varepsilon^3}{6}\cdot\frac{1}{N}\sum \Psi X_1X_2X_3$. So $d^3E/d\varepsilon^3 = \frac{1}{N}\sum \Psi X_1X_2X_3$, and the 1/6 is too much. (The code has no 1/6.)

3. The sum over absolute positions is not well defined. Use the ASR in two indices, with $X_1 = (X_1 - X_3) + X_3$ and $X_2 = (X_2 - X_3) + X_3$. A finite sum becomes $\sum_3 X_3 \cdot M_3$, with $M_3 = \sum_{12} \Psi_{123}(X_1 - X_3)(X_2 - X_3)$, proportional to the second-order force that the uniform strain puts on atom 3.

   - In an infinite crystal this sum depends on the image convention: $\tau \to \tau + R$ changes it by $R \cdot M_\tau$.
   - In a Bravais lattice $M = 0$ by inversion symmetry, so the sum vanishes, and the cubic energy is not captured at all.
   - `cubic_elastic` takes positions relative to atom 1 of the cell. With one atom per cell the first factor is then 0, so it returns exactly 0. For Ge (2 atoms) it prints $B_0' = 133$.

   The correct expression is Eq. (18), or Eq. (19) with f''' from Eq. (13).

4. The paper writes $B(V) = B(V_0)(1 + B_0' \Delta V/V_0)$. From $dB/dV = (dB/dP)(dP/dV) = -B_0' B/V$, the correct form is $B(V) = B(V_0)(1 - B_0' \Delta V/V_0)$. The comment in dynamical.f90 has the minus sign.

In FOCEX (as of 2026-10-27) the B_0' of `cubic_elastic` is only printed. The QHA overwrites it with $(1 + 6\gamma)/3$, with γ from Agne's $x = v_T/v_L$, and uses $B = B_0[1 - \eta(4 + 3\gamma)]$ for the bulk modulus along the QHA.

9 Finite temperature

9.1 The free energy

The Helmholtz free energy per cell is the static energy plus the vibrational free energy:

$$F(\eta, T) = E_0(\eta) + F_{\text{vib}}(\eta, T), \qquad E_0(\eta) \equiv W_{\text{rel}}(\eta) \ \ \text{(Eq. (10))}. \tag{23}$$

At temperature T and pressure P, the strain $\eta(T, P)$ minimises $F + PV(\eta)$ over the 6 strains. Then:

- $\alpha = \partial \ln V/\partial T|_P$;
- the isothermal elastic tensor is $C_T = V^{-1}\partial^2 F/\partial\eta\partial\eta$ at the minimum, and B_T follows from Eq. (18) with C_T in place of C;
- $C_V = -T\,\partial^2 F/\partial T^2$, $C_P = C_V + TV\alpha^2 B_T$, and $B_S = B_T C_P/C_V$.

For a cubic crystal $\eta = \varepsilon\mathbb{1}$, and F is a function of V alone. Both terms of F contribute to the pressure and to the bulk modulus:

$$P = -\frac{dE_0}{dV} - \frac{\partial F_{\text{vib}}}{\partial V} \equiv P_{\text{st}} + P_{\text{vib}}, \qquad B_T = V\frac{d^2E_0}{dV^2} + V\frac{\partial^2 F_{\text{vib}}}{\partial V^2} \equiv B_{\text{st}} + B_{\text{vib}}.$$

In a non-cubic crystal the thermal expansion is anisotropic, and the strain at T is not on the $T = 0$ pressure path. The minimisation has to be done over the 6 strains.

9.2 Force constants at a strained configuration

F_{vib} needs the phonons at strain η. Taylor's theorem applied to Eq. (1), about the displaced positions $X + S(p)$, gives the FCs there, exactly within the order of the expansion:

$$\begin{aligned} \Pi(p) &= \Pi + \Phi S + \tfrac{1}{2}\Psi SS + \tfrac{1}{6}\chi SSS, & \Phi(p) &= \Phi + \Psi S + \tfrac{1}{2}\chi SS, \\ \Psi(p) &= \Psi + \chi S, & \chi(p) &= \chi. \end{aligned} \tag{24}$$

At the relaxed p, $\Pi(p) = 0$. So the phonons at any strain need no new DFT data, only FC2–FC4 of the reference. With FC3 only, $\Phi(p)$ is linear in η.

9.3 Quasi-harmonic approximation

$F_{\rm vib} = k_B T \sum_{k\lambda} \ln\left[2\sinh(\hbar\omega_{k\lambda}/2k_BT)\right]$, with ω from $\Phi(p)$. For the cubic case, let $\gamma_{k\lambda} = -d\ln\omega_{k\lambda}/d\ln V$, $\epsilon_{k\lambda} = \hbar\omega(n+\frac{1}{2})$ the mode energy, $c_{k\lambda} = \partial\epsilon/\partial T$ its heat capacity, and $q_{k\lambda} = d\ln\gamma_{k\lambda}/d\ln V$. Then

$$P_{\rm vib} = \frac{1}{V}\sum_{k\lambda}\gamma\,\epsilon, \qquad B_{\rm vib} = \frac{1}{V}\sum_{k\lambda}\Big[\gamma\,\epsilon\,(1+\gamma-q) - T\gamma^2 c\Big]. \tag{25}$$

Derivation. $\epsilon = \hbar\omega\,(n(x)+\frac{1}{2})$ with $x = \hbar\omega/k_BT$. Differentiating, $\omega\,\partial\epsilon/\partial\omega = \epsilon - Tc$. Then differentiate $P_{\rm vib}$, using $d\omega/dV = -\gamma\omega/V$ and $d\gamma/dV = q\gamma/V$.

- To first order, $P_{\rm st} + P_{\rm vib} = 0$ gives $\Delta V/V_0 = \sum\gamma\epsilon/(V_0B_0)$, and $\alpha = \sum\gamma c/(V_0B_0)$. With the thermal Grüneisen parameter $\gamma_{\rm th} = \alpha V B_T/C_V$, $B_S = B_T(1+\alpha\gamma_{\rm th}T)$.
- $B_{\rm st}(V) = B_0(1 - B_0'\Delta V/V_0)$ to first order, with B_0' from Eq. (18).
- q needs the second derivative of ω^2 along the deformation (non-degenerate modes):

$$\frac{d^2\omega_\lambda^2}{d\lambda^2} = e_\lambda^\dagger D'' e_\lambda + 2\sum_{\mu\neq\lambda}\frac{|e_\mu^\dagger D' e_\lambda|^2}{\omega_\lambda^2 - \omega_\mu^2}.$$

  D'' comes from $d^2\Phi/d\lambda^2 = 2\Psi S_2 + \chi S_1 S_1$, where S_2 is the second-order relaxation (η_2, u_2^0). Without FC4, some value of q has to be assumed, and its effect tested.
- FOCEX's present $B = B_0[1-\eta(4+3\gamma)]$ is a single expression in place of both $B_{\rm st}$ and $B_{\rm vib}$.

9.4 With FC4: temperature-renormalised force constants

The quartic term changes the effective FC2 at *any* temperature, zero-point included, even at fixed strain:

$$K(p,T) = \Phi(p) + \tfrac{1}{2}\chi\,G, \qquad G_{ij} = \langle y_i y_j\rangle_K. \tag{26}$$

This is the paper's $K = \Phi + \Psi S + \frac{1}{2}\chi(SS + \langle yy\rangle)$, with $\Phi(p)$ from Eq. (24). The variational (self-consistent phonon) free energy, with V^{LD} expanded to fourth order about $X + S(p)$, is

$$F_{\rm vib} = F_h[K] + \left\langle V^{LD} - E_0 - \tfrac{1}{2}Kyy\right\rangle_K = F_h[K] - \tfrac{1}{8}\sum\chi\,G\,G. \tag{27}$$

$F_h[K]$ is the harmonic free energy of the frequencies of K. The second equality uses $\langle y\rangle = 0$ and $\langle yyyy\rangle = 3GG$ (Wick): $\frac{1}{2}\Phi G + \frac{1}{8}\chi GG - \frac{1}{2}KG = -\frac{1}{8}\chi GG$.

- One pass (G from $\Phi(p)$) is the first-order correction. Iterating to self-consistency is SCOP8's method.
- The second-order cubic term, $\propto \Psi^2$, is not in Eq. (27). Classically $\langle V_3^2\rangle \propto \Psi^2 G^3$ and $G \propto T$, so this term, $\sim -\langle V_3^2\rangle/2k_BT$, scales as T^2, like the quartic term χGG. It is of the same order and should be added for consistency.

C computed with $K(T)$ is not C_T. The paper suggests the elastic constants at T from A, Q, ϕ with $\Phi \to K(T)$. That is the curvature of the energy at fixed K. The isothermal tensor is the curvature of F, in which the phonon frequencies and populations change with the strain. In the QHA limit ($\chi = 0$), $C[K]$ is the static C at the expanded volume, i.e. $B_{\rm st}$, and $B_{\rm vib}$ of Eq. (25) is missing.

9.5 Procedure

1. **Static part.** From σ (input), FC2, FC3 and FC4: C, η_1, p_1, B_0, B_0' (Eq. (18)), and $E_0(\eta)$ to fourth order (Eq. (10)).
2. **Vibrational part.** At trial strains η, $\Phi(p)$ from Eq. (24), then F_{vib}: QHA, or Eq. (27) with FC4.
3. **Equilibrium.** Minimise $F + PV$ over η at each T.
   - Cubic: one variable; fit $F(V)$ on 5–9 volumes.
   - Otherwise: Newton steps in the 6 strains, with $\partial F_{\mathrm{vib}}/\partial\eta$ from the tensor Grüneisen parameters $-\partial \ln\omega/\partial\eta_{\alpha\beta}$ (FC3), or numerically.
4. **Output.** $V(T)$, $\alpha(T)$, C_T and $B_T(T)$ from the curvature at the minimum, C_V, C_P, B_S.
5. **Checks.** Eq. (25) against the numerical curvature (QHA, cubic); B_0' of Eq. (18) against dB/dP along the $T = 0$ path.

Test before coding. On the exact model (`FOCEX/REFERENCE/exact_model`) the energy is known exactly at any configuration.

- Relax cell and atoms numerically at a few pressures.
- Compare B, dB/dP and $C_{\alpha\beta\gamma\delta\mu\nu}$ with Eqs. (18) and (21).
- Use CaF_2 ($u_1^0 = 0$ by symmetry), wurtzite ZnO (free internal parameter) and HfO_2 (low symmetry).
- The model's reference is stressed. Test at the reference pressure ($P_0 \neq 0$; the CaF_2 stress is hydrostatic), or about a relaxed state found first.

References

[1] M. Born and K. Huang, *Dynamical Theory of Crystal Lattices* (Oxford University Press, 1954), Ch. V, §§26–27.

[2] D. C. Wallace, *Thermodynamics of Crystals* (Wiley, 1972; Dover, 1998).